

Test Plan: one missed verto.ping ends the call

signalwire-js #16 (server) and #17 (SDK). Rig: this repository, SDK @signalwire/js 4.0.0-rc.3. Page: README reproduction 8 / test-plan run I.

The question

On 2026-09-25 a signal-only reconnect covered a server `verto.ping`. No ping came on the new socket, and one interval (46s) later the server sent `call.left` + `verto.bye`. Is **one unanswered ping** enough to end a call, and does a reconnect lose it?

Setup (once)

- 1. `node serve.mjs`, open `https://localhost:8443/` (no flags).
- 2. Browser A and browser B (or incognito), same v4 SAT and same `/public/<room>`.
- 3. DevTools on A: Verbose on, Preserve log on.
- 4. Join both. Wait for **two** pings so the `verto.ping` row shows a countdown.
- 5. All actions are on **A**. B is the control and should stay in the call.

Every attempt

- 1. Reload A, rejoin, wait for the countdown.
- 2. Press **one** button (below).
- 3. Wait **two ping intervals** (about 95s).
- 4. Record: bye? how long after the action? `cause` / `causeCode`?
- 5. Save A's console (right-click, Save as) under the log name below.

The four variants

Button on A	What it does	If the call ends, it means	Log name
Drop socket as next ping lands	Closes the socket before the SDK sees the ping; the pong goes out on a CLOSING socket	The 09-25 sequence reproduces on demand	<code>run-I-drop-on-ping.log</code>
Drop socket just before next ping	Closes 300ms before the ping is due; the server sends it into the gap	A ping sent during a reconnect is never redelivered	<code>run-I-drop-before-ping.log</code>
Swallow next pong	Discards the pong, socket stays open, no reconnect	One missed pong is fatal on its own; reconnects are just one way to lose it	<code>run-I-swallow-pong.log</code>
Drop socket midway between pings (control)	Plain reconnect with no ping in the gap	The ping is not the cause (control failed)	<code>run-I-control-midway.log</code>

Results (3 attempts each)

Variant	Attempt 1: bye? / secs / cause	Attempt 2	Attempt 3
Drop as ping lands			
Drop just before ping			
Swallow next pong			
Control: drop midway			

Log lines that matter (on A)

- `verto.pong ... sent on WebSocket #k (CLOSING)` in red: answer was lost
- `SWALLOWED` the `verto.pong`: swallow variant armed and fired
- `... still unanswered when WebSocket #k closed`
- `verto.ping overdue`: no ping on the new socket
- `call.left` frame / `verto.bye` frame: note the cause
- `signal-only reconnect` must appear for drop variants

Discard the attempt if

- the log says `full reconnect` (peer connection dropped too)
- B also drops, or the room auto-ends because A left it alone
- the countdown was not showing when the button was pressed

If drop variants never end the call

A client close sends 1000; a real drop is 1006. Repeat with a ~1s wifi toggle timed off the countdown.